

Database Concepts -- Guided Notes

Unit tp-5.4 | CompTIA Network+ N10-009 Obj. FC0-U71 5.4 | N10-008 FC0-U71 5.4

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

A relational database stores data in _____ with rows and columns. The relationship between tables is defined using _____ keys that reference _____ keys in other tables. This structure prevents data _____ and ensures that related data stays consistent.

Question 2

ACID properties ensure reliable database transactions. A = _____ (all steps complete or none do), C = _____ (data remains valid), I = _____ (transactions do not interfere with each other), D = _____ (once committed, data persists even after a crash).

Question 3

A database _____ is a saved SQL query treated as a virtual table. It does not store data itself -- it runs the query dynamically. A database _____ is code stored in the database and executed on demand, allowing complex logic to run _____ to the data without round trips to the application server.

Question 4

An index in a database works like a book index -- it allows the DBMS to find records _____ without scanning every row. Creating an index on a frequently queried column speeds up _____ operations but slightly slows _____ and _____ operations because the index must be maintained when data changes.

Question 5

Database _____ is the process of copying data to another location to protect against data loss. _____ backups copy all data each time. _____ backups copy only data changed since the last full backup. _____ backups copy only data changed since the last backup of any type. The _____ (Recovery Point Objective) defines the maximum acceptable data loss window.

Question 6

A database _____ creates linked copies of the database on multiple servers for redundancy. A _____ replication model has one server that accepts writes and pushes changes to read-only _____. This improves _____ availability and _____ performance by distributing read queries.

Question 7

NoSQL databases trade strict _____ consistency for horizontal _____ (spreading data across many servers). The CAP theorem states that a distributed system can guarantee only _____ of these three: Consistency, Availability, and Partition Tolerance. Most NoSQL systems choose _____ and Partition Tolerance (AP) over strong consistency.

Question 8 -- Real World Application

REAL WORLD: An e-commerce website uses a single database server with no backup and no redundancy. One morning the server hard drive fails. The website is offline and all customer order history is gone. Describe three database infrastructure practices the company should implement to prevent this scenario from recurring.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-5-4-database-concepts/tp-5-4-database-concepts-answer-key.pdf
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